had it rough. In 1937, when Igor was two, his father, a high official in the Railway Ministry in Moscow, was entrapped by Stalin's Great Terror, arrested, and (less lucky than Landau) executed. His mother's life was spared; she was sent to prison and then exile, and Igor was reared by an aunt. (Such Stalin-era family tragedies were frightfully common among my Russian friends and colleagues.)

In the early 1960s, while I was studying physics as an undergraduate at Caltech, Igor was studying it as a graduate student at Moscow University.

In 1962, when I was preparing to go to Princeton for graduate study and do general relativity research with John Wheeler, one of my Caltech professors warned me against this course: General relativity has little relevance for the real Universe, he warned; one should look elsewhere for interesting physics challenges. (This was the era of widespread skepticism about black holes and lack of interest in them.) At this same time, in Moscow, Igor was completing his kandidat degree (Ph.D.) with a specialty in general relativity, and his wife, Nora, also a physicist, was being warned by friends that relativity was a backwater with no relevance to the real Universe. Her husband, for the sake of his career, should leave it.

While I was ignoring these warnings and pushing onward to Princeton, Nora, worried by the warnings, seized an opportunity at a physics conference in Estonia to get advice from the famous physicist Yakov Borisovich Zel'dovich. She sought Zel'dovich out and asked whether he thought general relativity was of any importance. Zel'dovich, in his dynamic, forceful way, replied that relativity was going to become extremely important for astrophysics research. Nora then described an idea on which her husband was working, the idea that the implosion of a star to form a black hole might be similar to the big-bang origin of our Universe, but with time turned around and run backward.[3] As Nora spoke, Zel'dovich became more and more excited. He himself had developed the same idea and was exploring it.

A few days later, Zel'dovich barged into an office that Igor Novikov shared with many other students at Moscow University's Shternberg Astronomical Institute, and began grilling Novikov about his research. Though their ideas were similar, their research methods were completely different. Novikov, already a great expert in relativity, had used an elegant mathematical calculation to demonstrate the similarity between the big bang and stellar implosion. Zel'dovich, who knew hardly any relativity, had